

Marketing Attribution Using SQL

Data Analysis using SQL

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Agenda

1. Getting Familiar With CoolTshirts
2. User Journey
3. Data Analysis of Campaigns
4. Reinvestment on Campaigns
5. SQL Queries

1.

**Getting familiar with
CoolTshirts**

GETTING FAMILIAR

CoolTshirts is using 8 campaigns which 2 of them are retargeting campaigns. While the first 6 campaigns are leading users to the landing page, the latter 2 are leading users directly to the checkout page.

All sources are part of a user journey and are bound to a specific "user_id" which reflects a unique "timestamp".

Those campaigns are;

- **Weekly Newsletter** and **Retargeting Campaign** provide sources through email marketing.
- Whereas **Paid Search** provides users from paid search, **CoolTshirt Search** campaign is the source of a direct search.
- **Retargeting Ad** also provides users from a social media channel which helps users to continue their journey and directs them to their before-visited page.
- **Getting To Know CoolTshirts** and **Ten Crazy CoolTshirts Facts** are sources of online article marketing. **"Interview With Cool T-shirts Facts"** can also be considered as part of online article marketing.

GETTING FAMILIAR

Here you see all campaigns and their sources created to draw user interest to the landing page of CoolTshirts.

Sources	Campaign Names
Google	Cool T-Shirt Search
NYTimes	Getting To Know CoolTshirts NYTimes
Medium	Interview With CoolTshirts Founder
Buzzfeed	Ten Crazy CoolTshirts Facts
Google	Paid Search
E-Mail	Weekly Newsletter
Facebook	Retargeting Ad
E-Mail	Retargeting Campaign

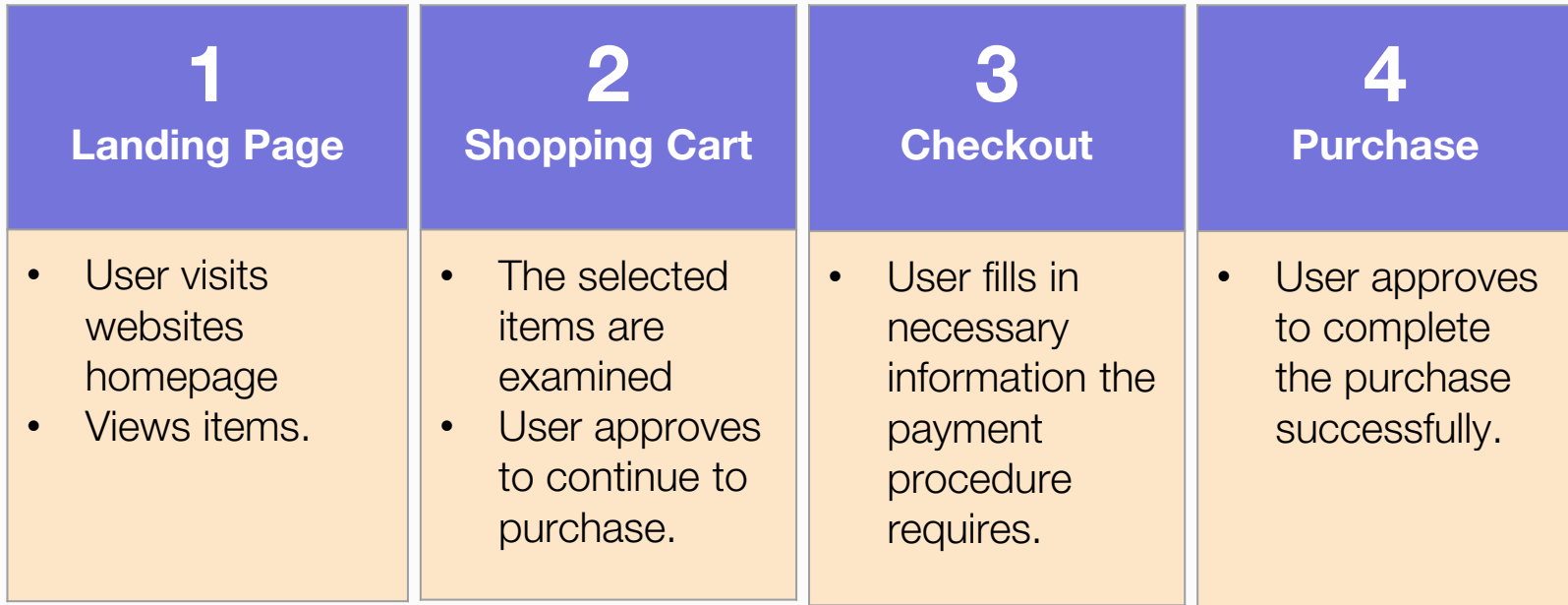
"utm_campaign" and "utm_source" are snippets attached parameters which addresses the source and kind of traffic. While the former makes it possible to track which campaign it was acquired from, the latter indicates the medium.

2.

User Journey

USER JOURNEY

The pages and basic actions user goes through in order to complete the purchase.



USER JOURNEY

While there are 8 campaigns created, only 6 sources are being used to draw users attention. You can see the two sources used twice.

Sources	Campaign Names
Google	Cool T-Shirt Search
NYTimes	Getting To Know CoolTshirts NYTimes
Medium	Interview With CoolTshirts Founder
Buzzfeed	Ten Crazy CoolTshirts Facts
Google	Paid Search
E-Mail	Weekly Newsletter
Facebook	Retargeting Ad
E-Mail	Retargeting Campaign

USER JOURNEY

And while the first four are campaigns aimed for first touch of the user on the landing page...

Sources	Campaigns Names
Google	Cool T-Shirt Search
NYTimes	Getting To Know CoolTshirts NYTimes
Medium	Interview With CoolTshirts Founder
Buzzfeed	Ten Crazy CoolTshirts Facts
Google	Paid Search
E-Mail	Weekly Newsletter
Facebook	Retargeting Ad
E-Mail	Retargeting Campaign



Intended to draw for first touch

USER JOURNEY

... these four are actions intended to remind users of their former interest in CoolTShirts.

Sources	Campaigns Names
Google	Cool T-Shirt Search
NYTimes	Getting To Know CoolTshirts NYTimes
Medium	Interview With CoolTshirts Founder
Buzzfeed	Ten Crazy CoolTshirts Facts
Google	Paid Search
E-Mail	Weekly Newsletter
Facebook	Retargeting Ad
E-Mail	Retargeting Campaign



Intended to remind user of their shown interest.

USER JOURNEY

Here are the numbers each first touch campaign is responsible for on the landing page.

Sources	Campaigns Names	Page Views
Google	Cool T-Shirt Search	169
NYTimes	Getting To Know CoolTshirts NYTimes	612
Medium	Interview With CoolTshirts Founder	622
Buzzfeed	Ten Crazy CoolTshirts Facts	576
Google	Paid Search	
E-Mail	Weekly Newsletter	
Facebook	Retargeting Ad	
E-Mail	Retargeting Campaign	

USER JOURNEY

The number of page views on the shopping cart page are the same numbers from the landing page since this page is only reached through the former.

Sources	Campaigns Names	Page Views	Page Name
Google	Cool T-Shirt Search	169	2 - Shopping Cart
NYTimes	Getting To Know CoolTshirts NYTimes	612	2 - Shopping Cart
Medium	Interview With CoolTshirts Founder	622	2 - Shopping Cart
Buzzfeed	Ten Crazy CoolTshirts Facts	576	2 - Shopping Cart
Google	Paid Search		
E-Mail	Weekly Newsletter		
Facebook	Retargeting Ad		
E-Mail	Retargeting Campaign		

USER JOURNEY

The number of page views on the shopping cart page are the same numbers from the landing page since this page is only reached through the former.

Sources	Campaigns Names	Page Views	Page Name
Google	Cool T-Shirt Search	7	3 - Checkout
NYTimes	Getting To Know CoolTshirts NYTimes	41	3 - Checkout
Medium	Interview With CoolTshirts Founder	21	3 - Checkout
Buzzfeed	Ten Crazy CoolTshirts Facts	32	3 - Checkout
Google	Paid Search	179	3 - Checkout
E-Mail	Weekly Newsletter	450	3 - Checkout
Facebook	Retargeting Ad	445	3 - Checkout
E-Mail	Retargeting Campaign	426	3 - Checkout

USER JOURNEY

The number of page views on the shopping cart page are the same numbers from the landing page since this page is only reached through the former.

Sources	Campaigns Names	Page Views	Page Name
Google	Cool T-Shirt Search	2	4 – Purchase
NYTimes	Getting To Know CoolTshirts NYTimes	9	4 – Purchase
Medium	Interview With CoolTshirts Founder	7	4 – Purchase
Buzzfeed	Ten Crazy CoolTshirts Facts	9	4 – Purchase
Google	Paid Search	52	4 – Purchase
E-Mail	Weekly Newsletter	115	4 – Purchase
Facebook	Retargeting Ad	54	4 – Purchase
E-Mail	Retargeting Campaign	154	4 – Purchase

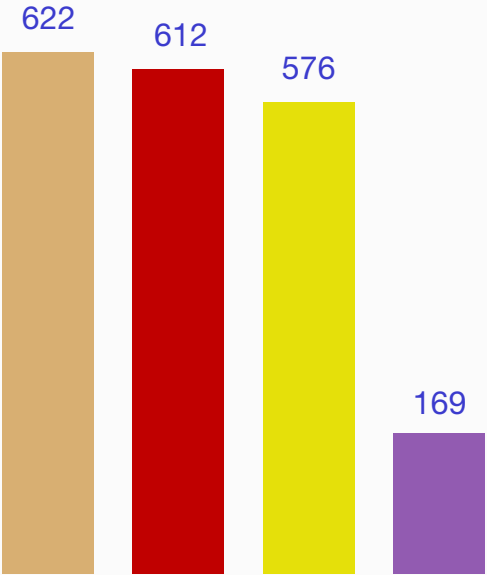
3.

Campaign Data Analysis

CAMPAIGN DATA

Top two campaigns that led users to discover the website are from **Medium** and **NYTimes**.

Top 2 campaigns that led users initially discover the website
Medium – Interview With CoolTshirts
NYTimes – Getting To Know CoolTshirts



- ten-crazy-cooltshirts-facts
- getting-to-know-cooltshirts
- interview-with-cooltshirts-founder
- cool-tshirts-search

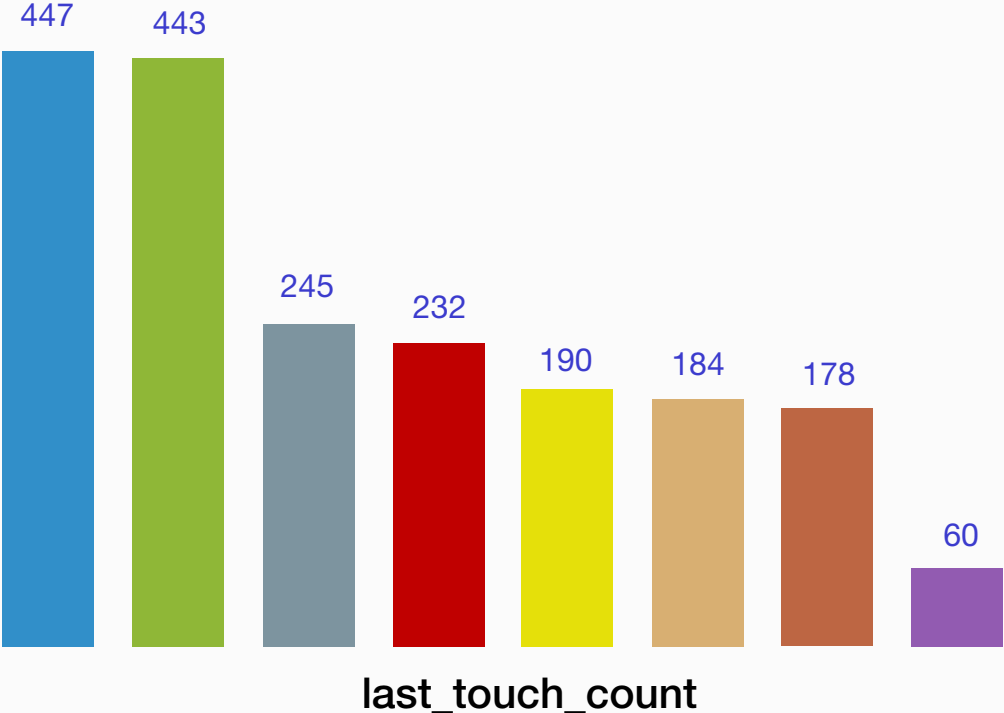
first_touch_count

CAMPAIGN DATA

Most purchases came from the **Weekly Newsletter** email campaign and **Facebook Retargeting Ad's**.

Top 2 conversion campaigns
Email – Weekly Newsletter
Facebook – Retargeting Ads

- weekly-newsletter
- retargeting-ad
- retargeting-campaign
- paid-search
- ten-crazy-cooltshirts-facts
- getting-to-know-cooltshirts
- interview-with-cooltshirts-founder
- cool-tshirts-search

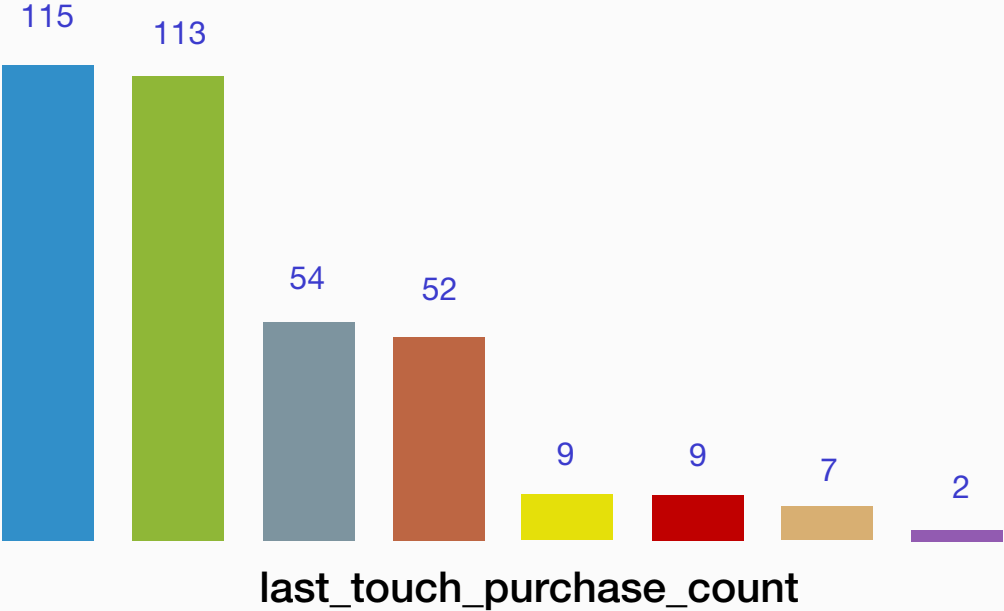


CAMPAIGN DATA

According to the data recieved, a 361 visitors have made a purchase.

Page Name	Purchases
4 – Purchase	361

- weekly-newsletter
- retargeting-ad
- retargeting-campaign
- paid-search
- ten-crazy-cooltshirts-facts
- getting-to-know-cooltshirts
- interview-with-cooltshirts-founder
- cool-tshirts-search



4.

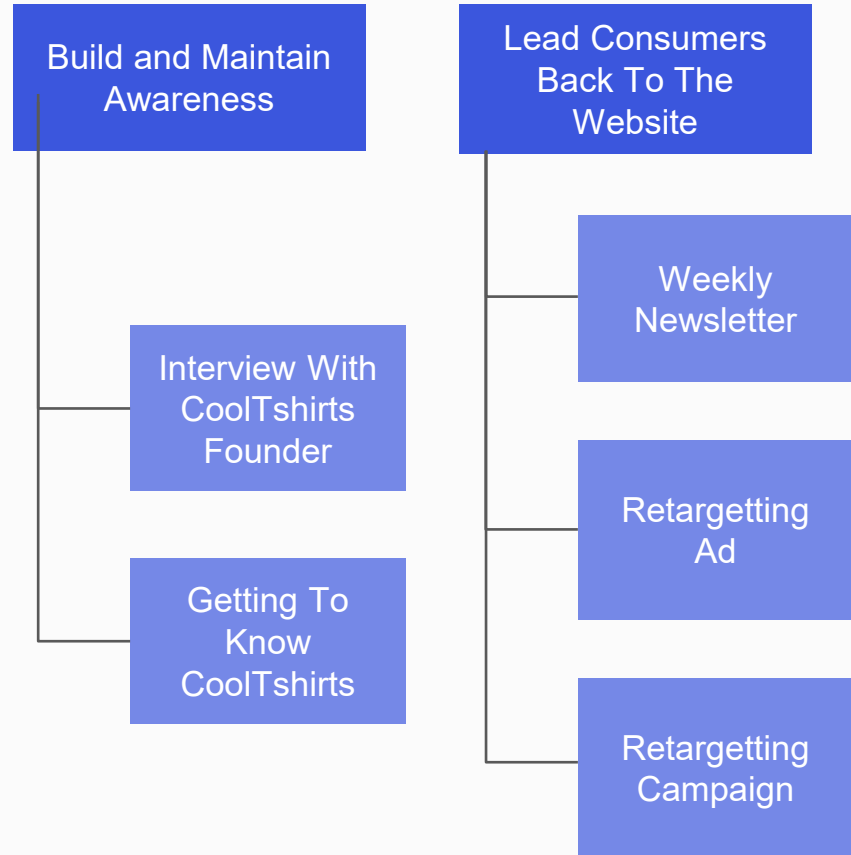
Reinvestment on Campaigns

CAMPAIGN DATA

With a limited budget, it would be advised to make use of a mixture of both **first touch** and **last touch** achieved campaigns.

Taking both **retargeting campaigns** and **advertisement campaigns**, the first five reinvestment advised campaigns would be;

- *Weekly Newsletter*
- *Retargeting Ad*
- *Retargeting Campaign*
- *Interview With CoolTshirts Founder*
- *Getting To Know CoolTshirts*



5.

Appendix SQL Queries

SQL QUERIES

SQL queries that belong to slides are explained below. Please note that some tables have been altered to explain data accordingly.

The query with the results for campaigns and sources. **Slide 5**

```
SELECT utm_campaign, utm_source  
FROM page_visits;
```

The query which gives the results for number of visitor making a purchase. **Slide 18**

```
SELECT DISTINCT page_name, COUNT(user_id)  
FROM page_visits pv  
WHERE pv.page_name = '4 – purchase'  
GROUP BY 1  
ORDER BY 1;
```

SQL QUERIES

The query describes the **first_touch** temporary table first, then describes the **ft_attr** which in it defines **first_touch** as **ft** and **page_visits** as **pv**. It also joins both tables by claiming **ft.user_id** and **pv.user_id** as an equation to generate the combination. And finally, it calls for the **ft_attr.utm_source** and **ft_attr.utm_campaign** columns, where also the integer count the number of each user id these columns while also binding both tables.

```
WITH last_touch AS (  
    SELECT user_id, MAX(timestamp) AS last_touch_at  
    FROM page_visits  
    GROUP BY 1  
)  
lt_attr AS (  
    SELECT lt.user_id, lt.first_touch_at, pv.utm_source, pv.utm_campaign  
    FROM last_touch ft  
    JOIN page_visits pv  
    ON lt.user_id = pv.user_id  
    AND lt.last_touch_at = pv.timestamp  
)  
SELECT lt_attr.utm_source, lt_attr.utm_campaign, COUNT(*)  
FROM lt_attr  
GROUP BY 1, 2  
ORDER BY 3  
DESC;
```

SQL QUERIES

The query which gives the number of purchases made from each campaign.

In the case of **last_touch**, there are no **first_touches** because users have to pass through the **Checkout** page before arriving at **Purchase**.

Slide 14

```
WITH last_touch AS (  
  SELECT user_id, MAX(timestamp) AS last_touch_at  
  FROM page_visits pv  
  WHERE pv.page_name = '4 - purchase'  
  GROUP BY 1),  
lt_attr AS (  
  SELECT lt.user_id, lt.last_touch_at, pv.utm_source, pv.utm_campaign  
  FROM last_touch lt  
  JOIN page_visits pv  
  ON lt.user_id = pv.user_id  
  AND lt.last_touch_at = pv.timestamp  
)  
SELECT lt_attr.utm_source, lt_attr.utm_campaign, COUNT(*)  
FROM lt_attr  
GROUP BY 1, 2  
ORDER BY 3  
DESC;
```


Thank you!